

LINUX COMMAND CHALLENGE

1. What is Server, OS, Database, Application and How They Are Related?

- Server: A computer or system that provides resources, data, services, or programs to other computers (clients) over a network.
- Operating System (OS): Software that manages hardware and software resources on a computer. Examples: Windows, Linux.
- Database: An organized collection of data stored electronically, managed by Database Management Systems (DBMS) like MySQL, Oracle.
- Application: Software designed to perform specific tasks for users, like web browsers, email clients, or custom business apps.
- Relationship: Servers run an OS, host databases and applications. Applications interact with databases to retrieve/store data, all operating on servers managed by the OS.

2. What is GUI and Command Line Interface?

- GUI (Graphical User Interface): A visual way to interact with computers using windows, icons, buttons, and menus (e.g., Windows desktop).
- Command Line Interface (CLI): A text-based interface where users type commands to perform tasks (e.g., Linux terminal).

3. What is Kali Linux and Why Is It Important in Cybersecurity?

- Kali Linux is a Debian-based Linux distribution designed for digital forensics and penetration testing.
- It comes pre-installed with numerous security tools for vulnerability assessment, ethical hacking, and penetration testing.
- Learning Kali Linux is vital for cybersecurity professionals to simulate attacks, find vulnerabilities, and secure systems.

4. List of 30 Kali Linux Commands with Uses:

Command	Use
ls	List directory contents
cd	Change directory
pwd	Print current directory
mkdir	Create directories
rm	Remove files or directories
cp	Copy files or directories
mv	Move or rename files or directories
cat	View file contents
nano	Text editor
grep	Search text in files
find	Search for files/directories
chmod	Change file permissions
chown	Change file owner
ifconfig	Show network interfaces
ping	Test network connectivity
netstat	Show network connections and stats
nmap	Network scanning and port discovery
airmon-ng	Enable monitor mode on wireless devices
aircrack-ng	Wireless network cracking
wireshark	Network protocol analyzer
tcpdump	Capture network packets

Command	Use
hydra	Password brute force attack tool
sqlmap	Automated SQL injection tool
msfconsole	Metasploit framework console
john	Password cracking tool
curl	Transfer data from or to a server
wget	Download files from the internet
tar	Archive files
ssh	Secure shell for remote login
sudo	Execute commands with root privileges
history	Show command history

5. Screenshot of Linux Command Challenge Completion:

